

Arc-Length Parameterized Interpolating Splines

Dafna K. Matsegora and Stephen M. Watt

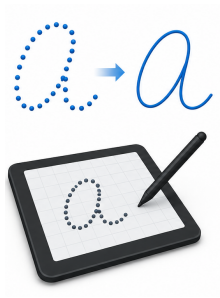
University of Waterloo, Cheriton School of Computer Science

July 20th, 2026

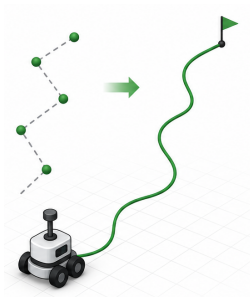
International Congress on Mathematical Software

Applications

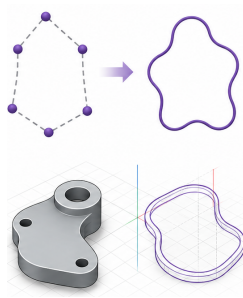
Many applications begin with discrete data points and require transforming them into smooth curves.



Handwriting Recognition



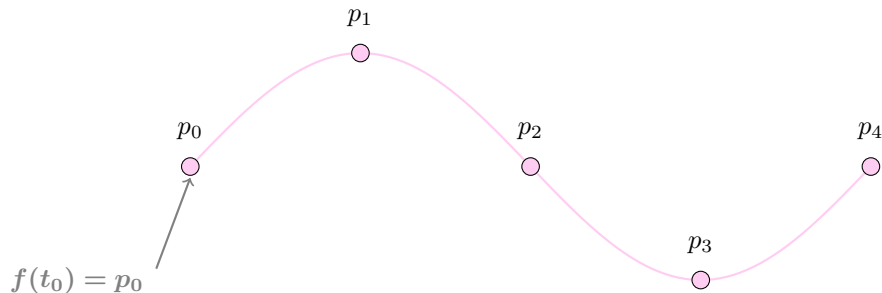
Robotics
Path Planning



CAD

Parameterization

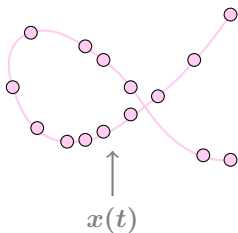
Parameterized, for example, by time or another monotonic parameter.



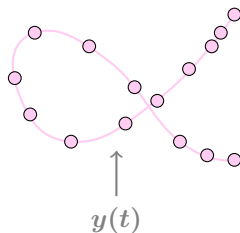
A Problem with Time-Based Data

What if the exact same shape is traversed at different speeds?

Tracing 1 (α):



Tracing 2 (α):



Same path, same shape, but time-based sampling gives very different point distributions.

Geometric Invariance

Parameterization should depend only on the geometry of the curve.

Desirable properties:

- Geometric invariance
- Monotonicity

Possible choices:

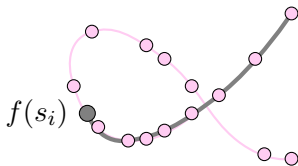
- Arc length
- Affine arc length
- Other geometric invariants

Arc length is the simplest and most natural choice.

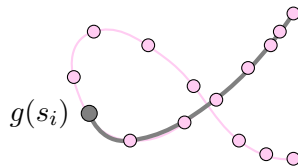
Arc-Length Parameterization

- s measures the distance travelled along the curve.
- Depends only on the geometry of the curve.
- Independent of velocity profile and point sampling.

Tracing 1



Tracing 2



$f(s_i) = g(s_i)$, with curves f and g such that $f \neq g$

The Problem

Given: Ordered data points $(p_0, p_1, \dots, p_n)$.

Find: Arc-length parameterized interpolating spline of desired type.

The Problem

Given: Ordered data points $(p_0, p_1, \dots, p_n)$.

Find: Arc-length parameterized interpolating spline of desired type.

Key requirements:

- Exact interpolation of the input data
- No assumptions on the sampling process
- Uniform parameterization by distance along the curve

Algorithm Overview

$$\{p_i\} \xrightarrow{\text{chord lengths}} \left\{s_i^{(0)}\right\} \xrightarrow{\text{piecewise linear interpolation}} f^{(0)}(s)$$

$\vdots$

$$f^{(k)}(s) \xrightarrow{\text{arc-length integration}} \left\{s_i^{(k+1)}\right\} \xrightarrow{\text{spline interpolation}} f^{(k+1)}(s)$$

$\vdots$

Spline Representation

Any spline can be used; for example, we use piecewise cubic splines with C^2 continuity at each knot:

$$f(s) = as^3 + bs^2 + cs + d$$

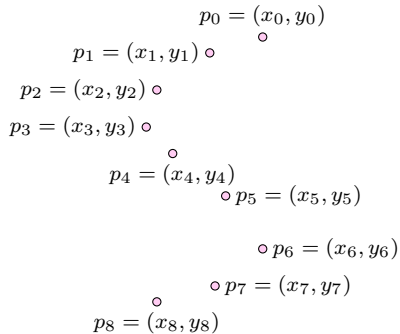
↓

f, f', f'' all continuous across knots

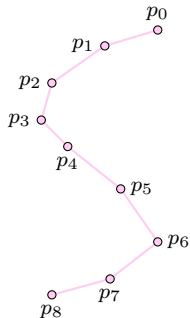
- Smooth curvature, not just smooth curves
- Efficient arc-length evaluation
- Higher-degree splines provide minimal accuracy gains for increased cost

Working example: S -shaped curve – Initialization

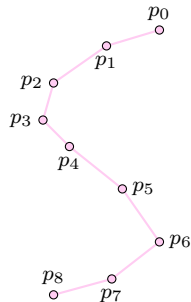
Input points



Chord connections

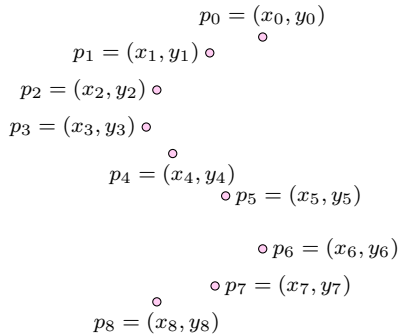


Initial Approximations

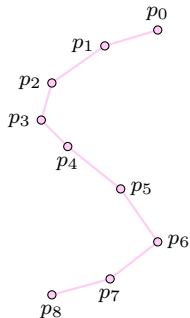


Working example: S -shaped curve – Initialization

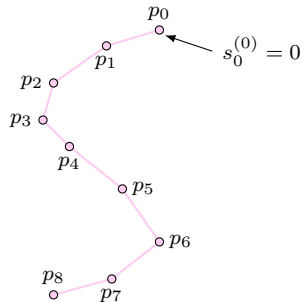
Input points



Chord connections

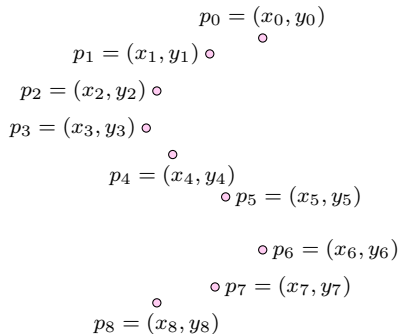


Initial Approximations

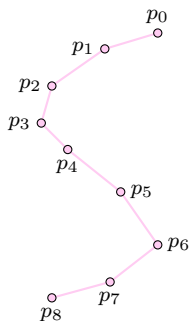


Working example: S -shaped curve – Initialization

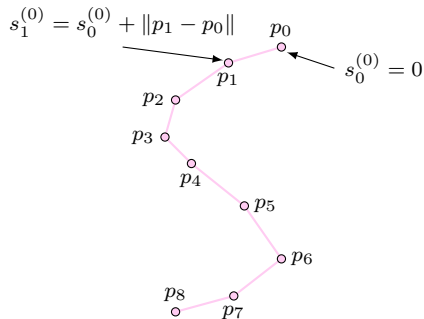
Input points



Chord connections

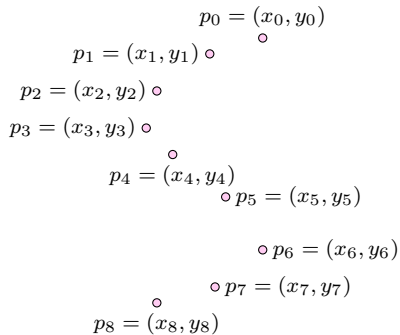


Initial Approximations

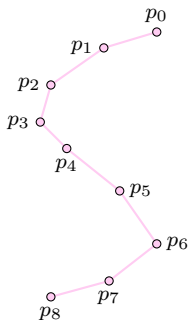


Working example: S -shaped curve – Initialization

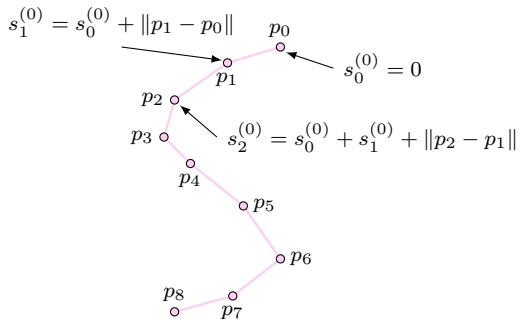
Input points



Chord connections

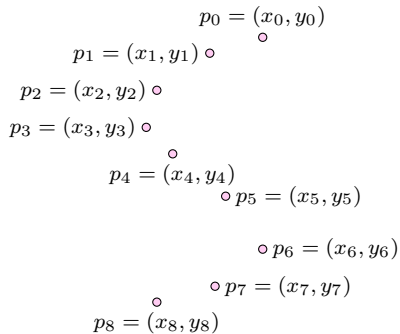


Initial Approximations

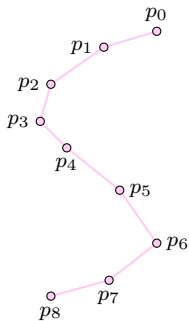


Working example: *S*-shaped curve – Initialization

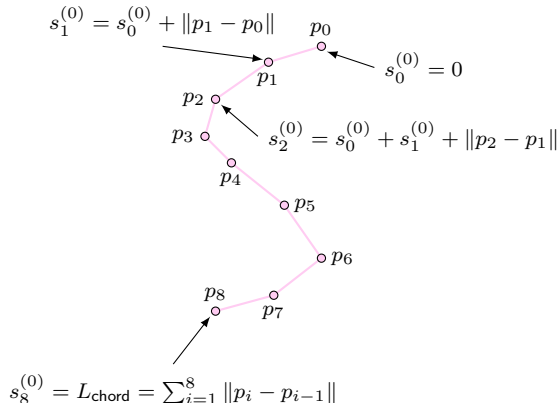
Input points



Chord connections

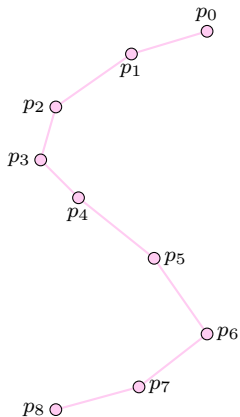


Initial Approximations



Working example: S -shaped curve – 1st Iteration

Initial Approximations



Using $s_i^{(0)}$, construct coordinate splines
 $x^{(1)}(s), y^{(1)}(s)$



Recompute the arc length of the i^{th} piece:

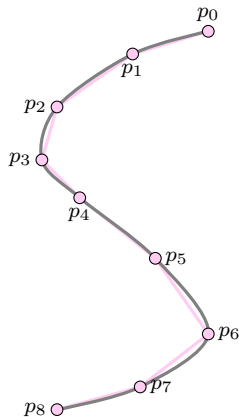
$$\ell_i^{(1)} = \int_{s_i^{(1)}}^{s_{i+1}^{(1)}} \sqrt{\left(\frac{dx^{(1)}}{ds}\right)^2 + \left(\frac{dy^{(1)}}{ds}\right)^2} ds$$



Update parameter values:

$$s_i^{(1)} = \sum_j \ell_j^{(1)}$$

First Spline Approximation



Working example: S -shaped curve – Termination

This process terminates under one of the following conditions:

- A fixed number of iterations is reached (when computational speed is prioritized).
- The normalized change in the parameters falls below a tolerance ε :

$$\frac{\max_i |s_i^{(k)} - s_i^{(k-1)}|}{s_n^{(k)}} < \varepsilon.$$

The normalization makes the stopping criterion independent of the curve's scale.

Working example: S -shaped curve – Final Output

- Arc-length parameter values

$$s_0^{(k)}, s_1^{(k)}, \dots, s_n^{(k)}$$

- Coefficients for each spline segment

$$(x_i(s)^{(k)}, y_i(s)^{(k)}), \quad i = 0, \dots, n - 1$$

Coefficients are not essential as output; they can be recomputed from the data points and the arc-length parameter values.

For dimension D

Given points, $p_i = (x_i^1, \dots, x_i^D)$, we compute D coordinate splines.

The length of the i -th spline segment becomes

$$\ell_i^{(k)} = \int_{s_i^{(k)}}^{s_{i+1}^{(k)}} \sqrt{\sum_{j=1}^D \left(\frac{dx^{j,(k)}}{ds} \right)^2} ds.$$

The parameter values are updated to be consistent with the arc lengths of the spline pieces.

Experimental Setup

Experiments

Cubic
interpolating splines

Periodic
boundary conditions

Sample sizes
 $n = 12, 24, \dots, 120$

Tolerance
 $\varepsilon = 10^{-1}, \dots, 10^{-6}$

Test Curves



Circle



Ellipse



Rose



Heart



Infinity



Hypotrochoid



Astroid



Lissajous

Some Experimental Results

Iterations to convergence over tolerances $\varepsilon \in \{10^{-1}, 10^{-2}, \dots, 10^{-6}\}$

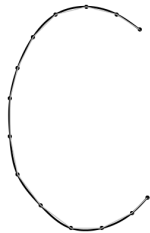
Shape	$n = 12$	$n = 24$	$n = 48$	$n = 72$	$n = 96$	$n = 120$
Ellipse	1 – 2	1 – 2	1 – 2	1 – 2	1 – 2	1 – 2
Rose	1 – 9	1 – 6	1 – 4	1 – 4	1 – 3	1 – 3
Heart	1 – 7	1 – 4	1 – 3	1 – 3	1 – 2	1 – 2
Infinity	1 – 5	1 – 3	1 – 2	1 – 2	1 – 2	1 – 2
Lissajous	1 – 9	1 – 8	1 – 6	1 – 4	1 – 3	1 – 3
Astroid	1 – 6	1 – 3	1 – 2	1 – 2	1 – 2	1 – 2
Hypotrochoid	1 – 4	1 – 3	1 – 2	1 – 2	1 – 2	1 – 2

- All curves converge within a small number of iterations
- More complex geometries require additional refinement
- Higher point counts generally improve convergence

A Cool Example: 'ICMS'



19 sample points
4 iterations



14 sample points
3 iterations



29 sample points
4 iterations



16 sample points
4 iterations

Convergence tolerance: $\epsilon = 10^{-6}$

Application

Handwritten Symbol $\rightarrow$ Arc-Length Parameterization $\rightarrow$ Legendre Series $\rightarrow$ Classification

Open problem

$$s^{(0)} \rightarrow s^{(1)} \rightarrow s^{(2)} \rightarrow \dots$$

Prove convergence

Empirically, all tested shapes so far have converged.